

Arrival · Supported route

FOUR QUICK QUESTIONS / POINT, SAY, SIGN, DRAW OR WRITE / YOU MAY PASS

- 1 Someone opens a bottle of perfume at the front. Can people at the back smell it after a while? Yes / No.
- 2 Two glasses of squash: one strong, one weak. Which one has more squash particles in the same amount of water? Strong / Weak.
- 3 In a gas, are the particles close together or far apart? Close / Far apart.
- 4 When you breathe in, where does oxygen get into your blood? Lungs / Stomach.

Nothing here assumes you have met diffusion before. The questions use everyday ideas about smells, drinks and breathing; an adult can read them to you or write your words. Choose one route. Point, say, sign, draw, type or write. You may pass or say not sure. Use the screen or paper; you do not need to do both.

Arrival - Stretch route

FOUR QUICK QUESTIONS / POINT, SAY, SIGN, DRAW OR WRITE / YOU MAY PASS

1. Some people press the bulk of a full moon. Explain what makes all the particles spread out when you're pushing them.
2. Two glasses of different temperatures of water. Which substance has more square particles give the same difference of water? Strong / Weak.
3. Why gas particles move faster than liquid / solid gas?
4. Why oxygen is in the air, the air is so close to the ground. Suggest why it goes that way and not the other.

- 2 Two glasses of different concentrations of the same substance are separated by a semipermeable membrane. Which side has more water? Strong / Weak.

- ### 3 Why do gas particles move faster than plastic/solid gas?

- 4 Why they pass the fire in, the air is so so of the lungs into the blood. Suggest why it goes that way and not the other.

Nothing here assumes you have met diffusion before. The questions use everyday ideas about smells, drinks and breathing; an adult can read them to you or write your words. Choose one route. Point, say, sign, draw, type or write. You may pass or say not sure. Use the screen or paper; you do not need to do both.

Make a prediction

Particles move randomly in both directions. Can there still be net movement from high to low concentration?

A No — random means no pattern

B Yes — more particles begin on the high side

C Only if every particle moves one way

Commit, then check

Diffusion knowledge organiser (1 of 2)

Objective: Explain diffusion and the factors that affect its rate.

KEY WORDS

Diffusion (Tier 3) — The net movement of particles from a higher concentration to a lower concentration.

Concentration (Tier 2) — How many particles of a substance there are in a given volume.

Concentration gradient (Tier 3) — The difference in concentration between two places.

Particles diffuse down it.

Net movement (Tier 3) — The overall movement once movement in both directions has been counted.

Passive (Tier 3) — Needing no energy from the cell; diffusion is passive.

Surface area (Tier 2) — The total area available for particles to cross.

Membrane (Tier 3) — A thin layer that particles must cross, such as a cell membrane.

Alveolus (Tier 3) — One of the tiny air sacs in the lungs where oxygen enters the blood.

Plural: alveoli.

Rate (Tier 2) — How fast something happens; here, how quickly particles move across.

SUCCESS CRITERIA

1. Define diffusion using net movement and concentration gradient.
2. Name three factors that change the rate of diffusion and say which way each one changes it.
3. Apply diffusion to a real example in the body and to the model shown, keeping the two apart.

Diffusion knowledge organiser (2 of 2)

KEY IDEAS

Particles move randomly all the time. Diffusion is the net result of that random movement, not a push.

Three factors raise the rate of diffusion: a steeper concentration gradient, a higher temperature and a larger surface area. A shorter distance to cross also raises it.

Diffusion needs no energy from the cell. Movement against a gradient using energy is active transport, a different process.

In the lungs, oxygen diffuses from the air in an alveolus into the blood; carbon dioxide diffuses the other way. The alveolus wall is one cell thick.

THE MODEL

A particle model of diffusion across a membrane. Particles are drawn far larger and fewer than the real thing, and the membrane is drawn as a line. This is an explanatory model, not a measurement of any real gas.

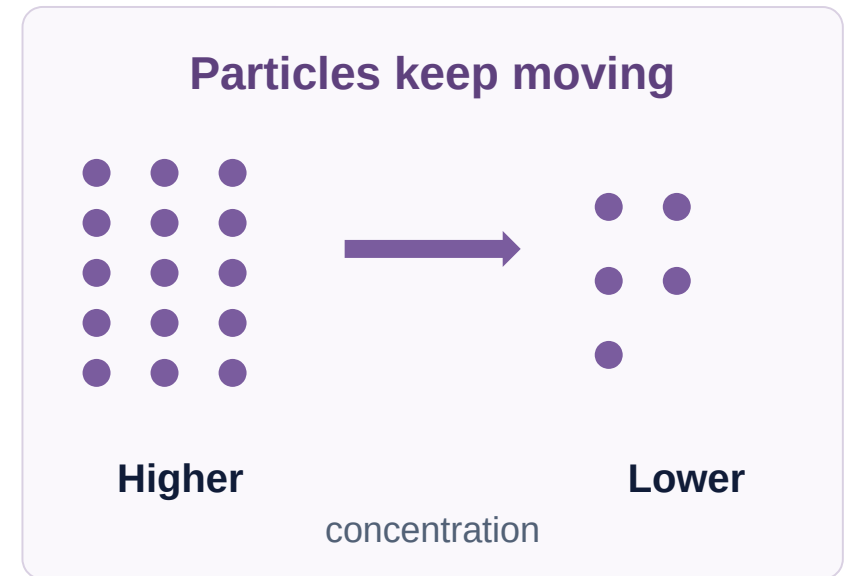
DIFFUSION OR NOT?

Diffusion: oxygen into the blood at the alveoli; carbon dioxide out of a respiring cell; a smell spreading across a room.

Not diffusion: glucose pumped into a root hair against its gradient (active transport); water across a partially permeable membrane (osmosis, a special case treated in the next lesson but one).

Watch the model

- 1 Individual particles move randomly in every direction.
- 2 More particles start in the high-concentration region, so more crossing events begin there.
- 3 Net movement is the overall difference between movements in opposite directions.



[View labelled diagram](#)

[Optional video](#)

Choose and justify

Which statement is scientifically accurate during diffusion?

A All particles move from high to low

B Particles stop when concentrations match

C Particles move randomly, with net movement down the gradient

D Low-concentration particles pull high-concentration particles

Give the deciding clue before checking.

Check the reasoning

Connect the science

- 1 A steeper concentration difference usually increases the net rate.
- 2 Higher temperature increases particle kinetic energy and movement rate.
- 3 A longer diffusion distance reduces the rate of exchange.

Illustrative crossings in equal volumes

Left → right: 8 Right → left: 2

Net left → right = $8 - 2 = 6$

At balance: $5 - 5 = 0$; motion continues.

Count both directions

Equal volumes: 12 particles on the left, 3 on the right. Predict the initial net direction.

Interval	Left → right	Right → left
Early	8	2
At balance	5	5

Find the net crossing count for each interval. What keeps happening at balance?

Need a different way to take part? (optional) During the sort, point, say, sign or show the adult an idea, a question or a change that would help you take part. You may pass or speak privately. The adult listens, says what can change and checks whether it helped.

Reveal shared reasoning

Your independent evidence

Equal-volume chambers have the same temperature and diffusion distance. A starts with 80 particles left, 20 right; B has 60 left, 40 right. Which has faster initial net diffusion? Explain using random motion and the gradient.

Optional support

Net movement = crossings one way minus crossings the other way. Start: “More particles move from ____ to ____ because...”

Make one precise correction to the same work after feedback.

[Open worksheet pages](#)[Model after first attempt](#)

Before you go · two small checks

CHOOSE ONE ROUTE, ONCE / SUPPORTED · STANDARD · STRETCH / YOU MAY PASS

- SUP 1** Diffusion is the net movement of particles from a higher concentration to a ____ concentration. lower / higher
- SUP 2** A steeper concentration gradient makes diffusion happen _____. faster / slower
- STA 1** Complete the definition: diffusion is the ____ movement of particles from a higher to a lower ____.
- STA 2** Name two factors that increase the rate of diffusion.
- STR 1** Explain why a thinner membrane increases the rate of diffusion. Use the word distance.
- STR 2** Explain why warming a gas increases the rate of diffusion. Use the words energy and particles.

Two small checks to finish. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once. Next lesson: Diffusion in the Lungs: Testing Breath.

Prediction check

Yes — more particles begin on the high side

[Return to Starter](#)

Check the deciding clue

Particles move randomly, with net movement down the gradient

Accurate. Use crossing counts to justify the net direction.

Compare your reason. Correct one step if needed.

[Return to We Do](#)

Our shared reasoning

Early: $8 - 2 = 6$ net left to right.

At balance: $5 - 5 = 0$ net.

Particles still move randomly and cross both ways. These are illustrative model counts.

[Return to shared task](#)

Compare after your first attempt

Chamber A has the faster initial net diffusion because its concentration difference is larger. Particles move randomly in both directions, but more particles begin on the left in A, so more left-to-right crossings occur at first. This produces a greater net movement down the concentration gradient.

Use the model to improve your explanation in your own words.

[Return to independent task](#)

Your lesson resources

Worksheet pages are stored inside this PowerPoint.

[Worksheet · page 1](#)[Worksheet · page 2](#)[Standard PDF](#)[Supported PDF](#)[Depth PDF](#)[Editable Word worksheet](#)[Open online lesson / model](#)

Named files are in this week's folder. Online lesson and video need internet.

[Optional concept video](#)

Worksheet · page 2

Continuation is optional.
Improve the same work.

MADE BY MATT / LAUNCH SCIENCE

W4L1 | Continue and improve

Use after the core questions when directed

3. Explain one reason a higher temperature can increase diffusion rate.

Optional transfer

Why must the two starting regions have equal volumes if particle counts alone are used to compare concentration?

Depth: name an assumption or limit, then say what evidence would strengthen your answer.

Improve the same work

One correction I made, and why:

My request for next time: _____

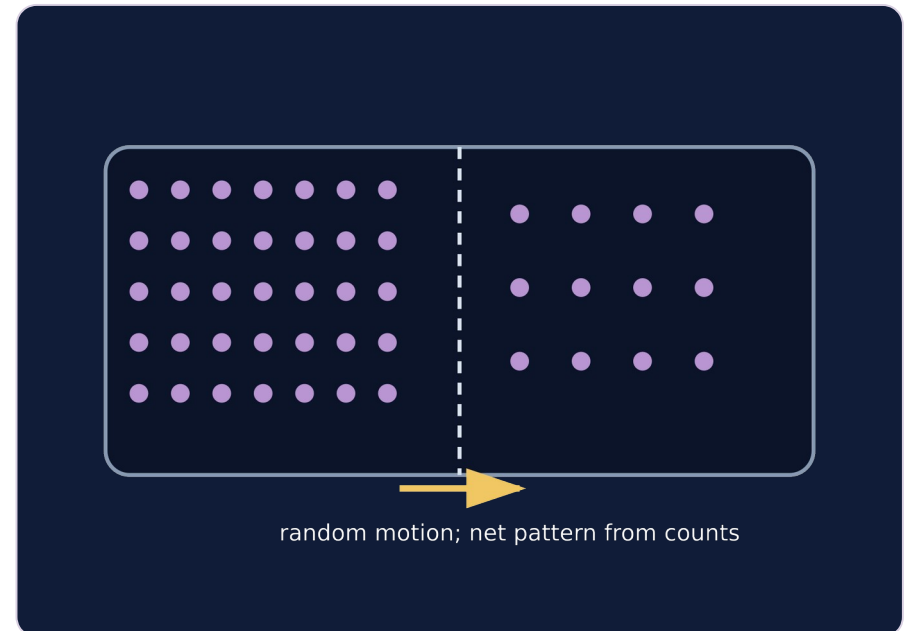
Return to task

Watch, pause and explain

Diffusion

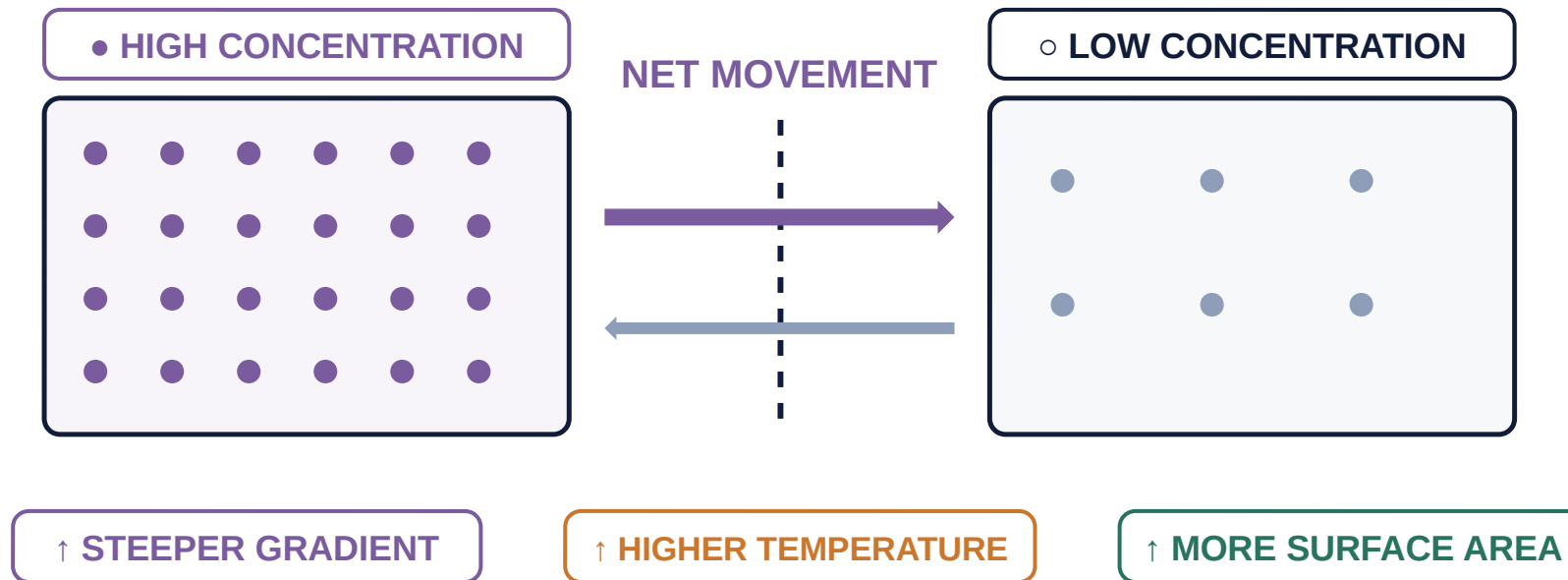
What keeps happening when there is no net movement?

[Play freesciencelessons video](#)



No connection? Use the still model and explain the same question.

Look closely at the model



A model illustration: use the stated data for calculations.

[Return to I Do](#)